Example 1

- At the annual Ponyville Harvest Festival, Applejack runs a game called "Barrel Toss," where players toss apples into one of five barrels lined up across the field. Applejack believes the barrels are evenly spaced and equally likely to receive apples if the game is fair.
- However, after watching several rounds, she suspects something might be off… maybe the barrels aren't truly equal. She records (**apple_toss**) the number of apples that landed in each barrel during the day (*barrel*) and wants to test whether they are uniformly distributed, as expected under fairness.

```
apple_toss |> n_pct(barrel)
```

```
   barrel    n (pct)
 Barrel 1 31 (15.5%)
 Barrel 2 29 (14.5%)
 Barrel 3 61 (30.5%)
 Barrel 4 44 (22.0%)
 Barrel 5 35 (17.5%)
```

```
apple_toss |> goodness_of_fit(outcome = barrel)
```

```
Chi-square goodness-of-fit test

Hypotheses:
H₀: Observed frequencies match expected proportions
H₁: Observed frequencies do not match expected proportions
Test statistic: χ²(4) = 17.1
p-value: 0.002
Conclusion: Reject the null hypothesis (p = 0.002 < α = 0.050)
```

Example 2

- Rarity tracks customer orders from her boutique (**rarity_orders**) across five product categories. She knows that sales typically follow this distribution:

Product	Proportion
Dresses	0.40
Capes	0.25
Hats	0.15
Scarves	0.10
Hoof Accessories	0.10

- This season, something feels off. Rarity reviews 200 orders from this season and wants to determine if the sales (*item_ordered*) are deviating from the historic pattern.

```
rarity_orders |> n_pct(item_ordered)
```

```
     item_ordered    n (pct)
            Cape 45 (22.5%)
           Dress 77 (38.5%)
             Hat 36 (18.0%)
 Hoof Accessories  17 (8.5%)
           Scarf 25 (12.5%)
```

```
rarity_orders |> goodness_of_fit(outcome = item_ordered,
                                 expected = c("Dress" = 0.40,
                                              "Cape" = 0.25,
                                              "Hat" = 0.15,
                                              "Scarf" = 0.10,
                                              "Hoof Accessories" = 0.10))
```

```
Chi-square goodness-of-fit test

Hypotheses:
H₀: Observed frequencies match expected proportions
H₁: Observed frequencies do not match expected proportions
Test statistic: χ²(4) = 3.51
p-value: 0.476
Conclusion: Fail to reject the null hypothesis (p = 0.476 ≥ α = 0.050)
```

Example 3

- Twilight Sparkle notices that her public library in Ponyville is visited by a variety of ponies: students, teachers, and general community members. She's curious whether the type of reading material ponies check out depends on their role in the community.
- She records each visitor's role (*role*; student, teacher, and community member) and the type of material they check out (*material*; spells, history, fiction, and science).
- Twilight wants to test whether the type of material checked out is independent of visitor role, or if certain roles are more likely to favor particular genres. Assume $\alpha = 0.05$.

```
twilight_library |> n_pct(material, role)
```

```
# A tibble: 4 × 4
  material Community  Student    Teacher
  <chr>    <chr>      <chr>      <chr>
1 Fiction  34 (56.7%) 9 (15.0%)  4 (6.7%)
2 History  11 (18.3%) 10 (16.7%) 25 (41.7%)
3 Science  9 (15.0%)  13 (21.7%) 8 (13.3%)
4 Spells   6 (10.0%)  28 (46.7%) 23 (38.3%)
```

```
twilight_library |> independence(role, material)
```

```
Chi-square test of independence

Hypotheses:
H₀: role and material are independent
H₁: role and material are not independent
Test statistic: χ²(6) = 57.55
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

Let's Practice! 1

Pinkie Pie records the weather on 120 festival days. If sunny, cloudy, rainy, and windy days are equally common, do her observations fit that pattern? Use $\alpha = 0.05$.

```
weather |> n_pct(weather)
```

```
 weather    n (pct)
  Cloudy 29 (24.2%)
   Rainy 30 (25.0%)
   Sunny 31 (25.8%)
   Windy 30 (25.0%)
```

```
weather |> goodness_of_fit(outcome = weather)
```

```
Chi-square goodness-of-fit test

Hypotheses:
H₀: Observed frequencies match expected proportions
H₁: Observed frequencies do not match expected proportions
Test statistic: χ²(3) = 0.07
p-value: 0.996
Conclusion: Fail to reject the null hypothesis (p = 0.996 ≥ α = 0.050)
```

Let's Practice! 2

Twilight records whether 120 earth ponies, pegasi, and unicorns use the train or a flight service for a trip. Is travel method associated with pony type? Use $\alpha = 0.05$.

```
travel |> n_pct(pony_type, travel_method)
```

```
# A tibble: 3 × 3
  pony_type  Flight     Train
  <chr>      <chr>      <chr>
1 Earth pony 12 (20.7%) 28 (45.2%)
2 Pegasus    30 (51.7%) 10 (16.1%)
3 Unicorn    16 (27.6%) 24 (38.7%)
```

```
travel |> independence(pony_type, travel_method)
```

```
Chi-square test of independence

Hypotheses:
H₀: pony_type and travel_method are independent
H₁: pony_type and travel_method are not independent
Test statistic: χ²(2) = 17.89
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

Let's Practice! 3

Fluttershy's animal clinic historically schedules 25% of visits in the morning, 50% in the afternoon, and 25% in the evening. Do 100 recent visits follow that pattern? Use $\alpha = 0.05$.

```
clinic |> n_pct(appointment_time)
```

```
 appointment_time    n (pct)
        Afternoon 30 (30.0%)
          Evening 20 (20.0%)
          Morning 50 (50.0%)
```

```
clinic |> goodness_of_fit(outcome = appointment_time,
                          expected = c("Morning" = 0.25,
                                       "Afternoon" = 0.50,
                                       "Evening" = 0.25))
```

```
Chi-square goodness-of-fit test

Hypotheses:
H₀: Observed frequencies match expected proportions
H₁: Observed frequencies do not match expected proportions
Test statistic: χ²(2) = 34
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

Let's Practice! 4

Fluttershy records the service chosen by 100 pet owners from Ponyville and Cloudsdale. Is service choice associated with town? Use $\alpha = 0.05$.

```
pet_care |> n_pct(town, service)
```

```
# A tibble: 2 × 4
  town       Checkup    Grooming   Training
  <chr>      <chr>      <chr>      <chr>
1 Cloudsdale 30 (75.0%) 10 (25.0%) 10 (50.0%)
2 Ponyville  10 (25.0%) 30 (75.0%) 10 (50.0%)
```

```
pet_care |> independence(town, service)
```

```
Chi-square test of independence

Hypotheses:
H₀: town and service are independent
H₁: town and service are not independent
Test statistic: χ²(2) = 20
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

Let's Practice! 5

Roseluck expects 40% roses, 30% daisies, 20% tulips, and 10% lilies. Do 100 recent flower purchases fit that distribution? Use $\alpha = 0.05$.

```
flowers |> n_pct(flower)
```

```
 flower    n (pct)
  Daisy 28 (28.0%)
   Lily 10 (10.0%)
   Rose 42 (42.0%)
  Tulip 20 (20.0%)
```

```
flowers |> goodness_of_fit(outcome = flower,
                           expected = c("Rose" = 0.40,
                                        "Daisy" = 0.30,
                                        "Tulip" = 0.20,
                                        "Lily" = 0.10))
```

```
Chi-square goodness-of-fit test

Hypotheses:
H₀: Observed frequencies match expected proportions
H₁: Observed frequencies do not match expected proportions
Test statistic: χ²(3) = 0.23
p-value: 0.972
Conclusion: Fail to reject the null hypothesis (p = 0.972 ≥ α = 0.050)
```

Let's Practice! 6

Applejack asks 120 ponies in three age groups whether they prefer an apple or a carrot. Is snack preference associated with age group? Use $\alpha = 0.05$.

```
snacks |> n_pct(age_group, snack)
```

```
# A tibble: 3 × 3
  age_group Apple      Carrot
  <chr>     <chr>      <chr>
1 Middle    18 (30.0%) 22 (36.7%)
2 Older     22 (36.7%) 18 (30.0%)
3 Young     20 (33.3%) 20 (33.3%)
```

```
snacks |> independence(age_group, snack)
```

```
Chi-square test of independence

Hypotheses:
H₀: age_group and snack are independent
H₁: age_group and snack are not independent
Test statistic: χ²(2) = 0.8
p-value: 0.670
Conclusion: Fail to reject the null hypothesis (p = 0.670 ≥ α = 0.050)
```